Chapter 3

Clinical and Social Perspectives on Pharmacy Services

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Learning Objectives

Objective 3.1 Position pharmacists in the healthcare system.
Objective 3.2 Describe the patient viewpoints on clinical pharmacy services.
Objective 3.3 Highlight the pharmacist perspective on clinical pharmacy services.
Objective 3.4 Describe the institutional and societal viewpoints on clinical pharmacy services.

OBJECTIVE 3.1. POSITION PHARMACISTS IN THE HEALTHCARE SYSTEM

During the past decades, the practice of pharmacists has lost two of the three main pillars of practice to the pharmaceutical industry: procuring and storing raw materials and producing drugs for each patient. The third pillar, the supply function of drugs, is not a viable lone pillar to build the future of pharmacy. The pharmacy profession, therefore, started the process to widen its scope to clinical pharmacy in the 1970s and to take a more active role in patient care.[1]

There is a difference in terminology across Europe, Great Britain, and North America regarding the concepts pharmacy practice and clinical pharmacy. In Great Britain, university departments that teach and research on pharmacy within healthcare in the broadest sense are often called *Pharmacy Practice*. In North America, Pharmacy Practice is the common name for the departments, but some of it is being renamed as *Pharmacotherapy and Outcomes Science or similar*. Countries in Northern Europe use the term *clinical pharmacy or social pharmacy*. One can, therefore, say that pharmacy practice is a broader concept than clinical pharmacy in that it both focuses on the clinical work of pharmacists with patients and on the social environment in which pharmacists work.

Knowledge of pharmacotherapy on its own is not enough to establish clinical pharmacy within the healthcare system. Understanding the broader field of social pharmacy is necessary for pharmacists to take on the responsibilities of providing patient care and clinical pharmacy services. Authors from Denmark and the United States define social pharmacy as a field dealing with the medical sector from the social scientific and humanistic perspectives.[2]

> *Topics relevant to social pharmacy consist of all the social factors that influence drug use, such as drug- and health-related beliefs, attitudes, rules, relationships, and processes.*[2]

The terms used to define social pharmacy vary between countries. The United Kingdom tends to include the clinical pharmacy, pharmaceutical policy, and social pharmacy within the term *pharmacy practice*. In the United States pharmacoepidemiology, pharmacoeconomics, and pharmaceutical policy—all considered as parts of social pharmacy in Europe—are seen as separate fields from pharmacy practice and clinical pharmacy.[3] The knowledge from social pharmacy about the function that various stakeholders have in the healthcare system, and knowledge of pharmacoepidemiology, pharmacoeconomics, and pharmaceutical policy, is needed for pharmacists to establish themselves as distinct clinical practitioners.

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Clinical pharmacy deals with caring for patients' pharmacotherapy and with rational drug use. Clinical pharmacy has been defined in various ways by professional organizations and academics across the world. The most used definitions are by the European Society of Clinical Pharmacy[4] and the American College of Clinical Pharmacy[5] (see Table 3.1).

When comparing these definitions, they overlap on three important issues.

Firstly, the focus of clinical pharmacy is on the pharmacist's role in patient care. Pharmacists' patient care services are expanding with prescribing and immunization services.

Secondly, clinical pharmacy has a focus on ensuring the rational and appropriate use of drugs. The World Health Organization defines the rational drugs use as follows:

> *The rational use of drugs requires that patients receive medications appropriate to their clinical needs, in doses that meet their requirements, for an adequate period and at the lowest cost to them and their community.*[6]

Thirdly, clinical pharmacy can be practiced in a wide variety of settings at different levels of care. Approaches of pharmacists to provide clinical services vary from setting to setting. In a tertiary level, care facility pharmacists could provide full-time clinical pharmacy service at the bedside. Advanced level of patient care services is provided by clinical pharmacists in collaboration with other healthcare professionals. These pharmacists are called clinical pharmacists as they are mainly providing direct patient care services. A community pharmacist spends the vast majority of their time on drug supply and other pharmacy management responsibilities. It does, however, not exclude provision of clinical services. All the cognitive services provided by a community pharmacist are clinical when they involve in direct patient care.

Community pharmacies and other primary care services are often the first point of call for both chronically ill patients and those seeking advice for the first time. Initiatives have been taken to involve pharmacists more actively in patient care away from the dispensing counter in pharmacies. An example is home care services offered by the Australian Home Medication Review service, where pharmacists make domiciliary visits to chronically ill patients and provide pharmaceutical care.[7]

In addition, the pharmacists should engage in research to generate new knowledge on how clinical pharmacy services advance human health and improve quality of life. It is important that pharmacists do research on the implementation and explore the impact of their services. Pharmacists provide clinical services to achieve definite health outcomes. They are well positioned in the society to cater to healthcare needs of patients and other consumers of healthcare.

TABLE 3.1 Definitions of Clinical Pharmacy

European Society of Clinical Pharmacy (ESCP) Clinical Pharmacy is a health specialty, which describes the activities and services of the clinical pharmacist to develop and promote the rational and appropriate use of medicinal products and devices. Clinical Pharmacy includes all the services performed by pharmacists practicing in hospitals, community pharmacies, nursing homes, home-based care services, clinics and any other setting where medicines are prescribed and used. The term "clinical" does not necessarily imply an activity implemented in a hospital setting. It describes that the type of activity is *related to the health of the patient(s).* This implies that community pharmacists and hospital pharmacists both can perform clinical pharmacy activities.[4]
The American College of Clinical Pharmacy (AACP)—Abridged Definition Clinical Pharmacy is that area of pharmacy concerned with the science and practice of rational medication use.[5]
The American College of Clinical Pharmacy (AACP)—Unabridged Definition Clinical Pharmacy is a health science discipline in which pharmacists provide patient care that optimizes medication therapy and promotes health, wellness, and disease prevention. The practice of clinical pharmacy embraces the philosophy of pharmaceutical care; it blends a caring orientation with specialized therapeutic knowledge, experience, and judgment for the purpose of ensuring optimal patient outcomes. As a discipline, clinical pharmacy also has an obligation to contribute to the generation of new knowledge that advances health and quality of life. Clinical pharmacists care for patients in all healthcare settings. They possess in-depth knowledge of medications that is integrated with a foundational understanding of the biomedical, pharmaceutical, sociobehavioral, and clinical sciences. To achieve desired therapeutic goals, the clinical pharmacist applies evidence-based therapeutic guidelines, evolving sciences, emerging technologies, and relevant legal, ethical, social, cultural, economic and professional principles. Accordingly, clinical pharmacists assume responsibility and accountability for managing medication therapy in direct patient-care settings, whether practicing independently or in consultation/collaboration with other healthcare professionals. Clinical pharmacist researchers generate, disseminate, and apply new knowledge that contributes to improved health and quality of life. Within the system of healthcare, clinical pharmacists are experts in the therapeutic use of medications. They routinely provide medication therapy evaluations and recommendations to patients and healthcare professionals. Clinical pharmacists are a primary source of scientifically valid information and advice regarding the safe, appropriate, and cost-effective use of medications.[5]

OBJECTIVE 3.2. DESCRIBE THE PATIENT VIEWPOINTS ON CLINICAL PHARMACY SERVICES

Clinical pharmacy emerged based on the need to improve pharmacotherapy of patients to maximize the health outcomes and to minimize the costs and risks associated with the irrational use of drugs. Pharmacists can take on a significant role in improving drug use and lowering the risk to patients.

Two challenges are important for pharmacists to consider. The first is that a high proportion of patients use many drugs concomitantly, some of them being unnecessary—also called polypharmacy. Another important challenge is that patients have their rationales and perceptions of drugs' use, which affects adherence, and therefore need to be understood and taken into account when providing drug information or counseling.

Polypharmacy and Inappropriate Drug Use

Treatment with drugs has many positive effects on the health and quality of life at all ages by preventing and treating various diseases. The health gains with drugs should, however, be greater than the troublesome side effects and other unwanted effects. Inappropriate prescribing occurs when potential risks are greater than the benefits of drug treatment. The concomitant use of many drugs is called *polypharmacy*—defined by different authors as taking a minimum of 4−10 drugs at the same time.[8] The number of drugs as such has no direct connection with neither pharmacodynamic or pharmacokinetic challenges, nor does it mean that some of the drugs are unnecessary. It is, however, proven that the number of drugs increases the risk for drug therapy problems, such as drug−drug interactions, side effects, inappropriate medications, or inappropriate dosages. It is also well known that old age affects the pharmacokinetics and efficacy of drugs because of physiological decline.[9]

A linear relationship exists between the number of drugs taken and the risk of having an adverse drug reaction (ADR), where each additional drug increases the risk of ADRs by 8.6% as per a study.[10] *Deprescribing* is defined as "the process of withdrawal of an inappropriate medication, supervised by a healthcare professional with the goal of managing polypharmacy and improving outcomes."[11] Scott et al. define deprescribing as "the systematic process of identifying and discontinuing drugs in instances in which existing or potential harms outweigh existing or potential benefits within the context of an individual patient's care goals, current level of functioning, life expectancy, values, and preferences."[12] Research has shown that doctors are reluctant to deprescribe drugs because of various reasons,[13] and here pharmacists are needed to support these efforts.

Rational use of drugs is at large about patients taking drugs appropriately to treat their clinical needs, in appropriate dosages. A related concept is the Goldilocks Principle, which means that the patient gets the right number of drugs—neither too much nor too little—just the right number and dose to treat the diseases.

The problems associated with polypharmacy and inappropriate drug use are not easy to solve. A systematic review and metaanalysis of whether pharmacist-led medication review can help to reduce hospital admissions and deaths in older people was in fact inconclusive. Pharmacist-led medication review suggested that interventions could improve knowledge and adherence and, in some studies, reduce the number of prescribed drugs, but had little, if any, effect on reducing mortality or hospital admission in older people.[14] The role of pharmacists in deprescribing should be reducing the number of (potentially) inappropriate medications and minimizing risks and maximizing outcomes of drug use.[15]

Nonadherence to Pharmacotherapy—The Patients' Perspective on Taking Drugs

The term adherence to medication is defined as "the process by which patients take their medications as prescribed."[16] Giving patients clear and concise instructions on how to take their drugs is no guarantee for that they will take it as instructed. In fact, roughly half of medications prescribed for long-term conditions are not taken appropriately by the patients. As a result, neither the patient nor the society has optimal outcomes from the drugs.[17,18]

Most would agree that our attitudes or understanding of any given subject affects our behavior. To explain nonadherence, one must take into account that all patients have their rationales and perception of drugs.[19] For pharmacists and other health personnel, it is important to develop an understanding of how patients' perspective on taking drugs directly or indirectly affects adherence. There is no single "solution" to rectify nonadherence, be it intentional or unintentional from the patients' perspective. Horne and Weinman[20] made a model on the beliefs about drugs, which has been used by many to study how patients are taking drugs. On one hand, patients have *necessity beliefs* on why they need to take drugs. On the other hand, patients have *concerns* about a range of potential adverse events from drug treatment. This balance, depending on whether concerns are stronger than the necessity or vice versa, directly affects adherence to drugs. So why do patients take less medicines and dosages than prescribed?

Intentional nonadherence occurs when patients make an active decision not to follow the instructions for a particular treatment. This behavior is best understood by patients' beliefs about drugs and the preferences people have for one kind of drug or treatment option compared with another. Intentional nonadherence is also influenced by previous experiences with a drug or disease, and this directly influences their motivation to start, carry on, or stop taking a drug.

Unintentional nonadherence, on the other hand, occurs for instance when patients have a desire to adhere to the treatment but are unable to do so. Unintentional nonadherence is often related to practical aspects of taking medicines: the information on how to take the drug is not understood, patients find it difficult to remember to take it, or it is difficult to administer.

A range of other factors additionally affect adherence. These are, for instance, the socioeconomic group one belongs to, cultural perceptions of illness and health, availability to healthcare services, or the disease itself. Pharmacists and other health personnel must tailor drug information and guidance on how to take drugs. They should openly discuss patients' intention to adhere to treatment. Research supports patient-centeredness in clinical decisions to improve adherence. Patients should be involved in the choice of drugs to clarify their responsibility and awareness of effects and side effects. For further reference, the NICE Medicines Adherence Guideline provides an overview of interventions to enhance adherence and insight into studies of patients' nonadherence.[17]

OBJECTIVE 3.3. HIGHLIGHT THE PHARMACIST PERSPECTIVE ON CLINICAL PHARMACY SERVICES

Pharmaceutical Care—A Central Concept in Patient Care

Pharmaceutical care as a concept has moved the pharmacy profession from primarily focusing on the product (the drug itself) to the patient's drug therapy and how it should be optimized for the individual patient.

The original definition of the concept of pharmaceutical care was published by Hepler and Strand in 1990 as follows "Pharmaceutical care is the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient's quality of life."[21] However, new definitions have emerged. The definition from a group in Minnesota, USA, which has been the most instrumental originally dating from 1998, is patient-centered and underlines that the practitioner (usually a pharmacist) is responsible for the patient's drug therapy:

> *Pharmaceutical care is a practice in which the practitioner takes responsibility for a patient's drug-related needs, and is held accountable for this commitment. In the course of this practice, responsible drug therapy is provided for the purpose of achieving positive patient outcomes.*[22]

A more recent definition by the Pharmaceutical Care Network Europe (PCNE), specifying pharmaceutical care as a service provided by pharmacists, states that:

> *Pharmaceutical Care is the pharmacist's contribution to the care of individuals in order to optimize medicines use and improve health outcomes.*[23]

According to Cipolle et al., the philosophy of pharmaceutical care practice consists of four elements. First, the social need to reduce drug-related morbidity and mortality associated with medication use. Second, the social need is met by the practitioners' responsibility to identify, resolve, and prevent drug therapy problems. Third, this service is provided in a patient-centered context, which means that the practitioner is working for the patient directly. The last part is caring, which means that the practitioner takes the time to assess, make a care plan, and follow up on the patient.[24]

In the United Kingdom or the United States, pharmacists often refer to providing *medication management* when optimizing patients' medication therapy[25] and not to the concept of *pharmaceutical care*. Interestingly, the new concept of Pharmacists' patient care process[26] has been introduced in the United States, which represents the care process delivered by pharmacists. It was developed by examining some key source documents on pharmaceutical care and medication therapy management.

Medication Reviews and Drug Therapy Problems

According to PCNE, a medication review is "a structured evaluation of patient's medication with the aim of optimizing drugs use and improving health outcomes. It entails detecting drug-related problems and recommending interventions."[27] Medication reviews require knowledge of pharmacotherapy and an ability to apply this knowledge to the individual patient. It is widely accepted that there are different levels of medication review depending on which information sources

are available and whether the patient is involved in the review process.[28,29] The levels of a medication review range from addressing technical issues based merely on prescription information to the advanced level of medication reviews. In medication reviews, details of the drug and the clinical conditions are discussed with the patient and a multiprofessional team.[28,29]

However, deep knowledge and skills in pharmacotherapy are not enough for a pharmacist to tackle the problems of inappropriate drug use. As Strand and Cipolle stated back in 1989:

> *Pharmacists must define their professional function from the perspective of the whole patient, the biopsychosocial sense, and not primarily from that of the pharmaceutical agent (e.g., gentamicin, theophylline) or the technical instrument employed (e.g., pharmacokinetic dosing, cholesterol measurement, blood pressure monitoring, nutritional protocol development).*[30]

The ability to work in multidisciplinary teams is also important for pharmacists to bring their knowledge and skills to bear on the patients' drug therapy and ultimately impact their quality of life.[31]

Cognitive Pharmaceutical Services

Pharmacist cognitive services are a range of healthcare-related activities provided by the pharmacist in all settings of care. Cognitive pharmaceutical services are strategies and actions to improve the patients' drug therapy by using pharmacists to the fullest extent of their training and education as medication experts.[32] Cognitive pharmaceutical services have been defined as follows:

> *Professional services provided by pharmacists, who use their skills and knowledge to take an active role in patient health, through effective interaction with both patients and other health professionals.*[33]

Examples of pharmacist cognitive services are within the realm of healthcare promotion, disease management and prevention, and services regarding drug therapy issues such as medication review. Fig. 3.1 shows how wide-ranging pharmacist cognitive services can be, but they can be roughly divided into activities related to health promotion, preventive and disease management, drug therapy issues, and prescribing.

Sometimes all pharmacists' services beyond dispensing drugs, have been referred to as pharmaceutical care service.[32,38] As seen in Fig. 3.1, pharmaceutical care is listed as one type of cognitive service. There are many types of services listed under the heading of Drug Therapy Issues in Fig. 3.1 that are related to and sometimes the components of pharmaceutical care, such as medication review, changing drugs/dosages, and patient education and monitoring. Many of these services can be provided in conjunction with dispensing in community pharmacies and then add a clinical dimension to the work of the pharmacist. However, pharmaceutical care is defined by Cipolle et al. as larger than all the other cognitive services because it states that the pharmacist takes the time to assess, make a care plan, and follow up on the patient. This service is, therefore, less suited in conjunction with dispensing because it encompasses reviewing and evaluating the patient's medications in a dialog with the patient and through this taking responsibility for the patient's whole medication therapy being appropriate, safe, and effective.

OBJECTIVE 3.4. DESCRIBE THE INSTITUTIONAL AND SOCIETAL VIEWPOINTS ON CLINICAL PHARMACY SERVICES

Clinical pharmacy services are often thought of as those where the pharmacist discusses medication therapy with the patient and other healthcare professionals. However, pharmacists also give structural support to health personnel. In hospitals they for example run drug information centers that primarily assist nurses and doctors with medication-related queries. Pharmacists also contribute to drug formularies in hospitals, with immense economic implications for the organization. Here the pharmacists contribute to rational drug prescribing and cost containment with implications to the institution and more broadly to the society.

There are evident challenges in society regarding the use of drugs, some of which will be highlighted here. Large proportions (1.4% in the OECD countries) of country's gross domestic product are spent on healthcare, and for most countries, medication expenditures constitute about one-fifth of total healthcare spending in OECD countries.[39] Taking drugs also involves risk of side effects and drug-related morbidity and mortality. In addition to incurring pain and loss, these risks have also been quantified into economic costs. As an example, it was found that in Germany the costs of drug-related morbidity could be as high as 816 million Euros in 2007.[40] Governments seek to contain these costs through a variety of measures.

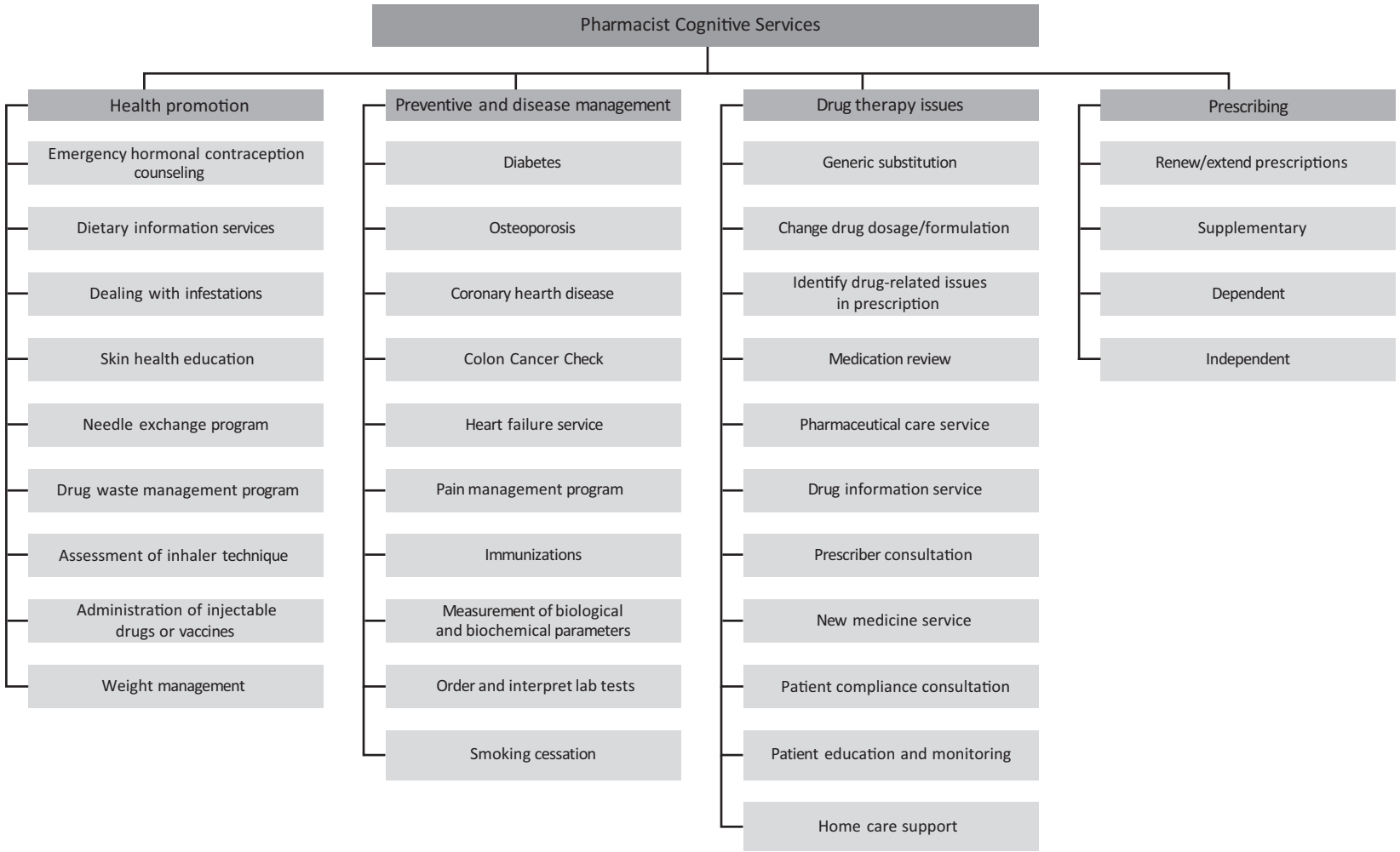


FIGURE 3.1 Examples of pharmacist cognitive services compiled from the cited articles.[34–37]

Barriers and Facilitators for Implementing Clinical Pharmacy Services

Providing additional clinical pharmacy services such as medication review and detailed counseling is not common within community pharmacies[36,38] or primary care.[37] It has been shown to be difficult to implement pharmacist services in community pharmacies beyond dispensing, probably because managing a pharmacy and dispensing drugs is a full-time job in its own right. Providing additional cognitive services needs more staff with limited or no dispensing roles. Pharmacies ought to have separate consulting rooms dedicated to the more intimate patient assessments such as to diagnose minor ailments, counseling sessions, address nonadherence behavior, or to administer vaccines, etc. Many pharmacies offer services such as monitoring body mass index, blood pressure, blood sugar, etc. Although barriers to new pharmacy services differ across countries,[41] they can be categorized into three domains: resources, attitude and opinion, and lack of education and skills.[42–44] It is likely that community pharmacies in different countries will differ even more in the years to come, depending on strategies chosen on which roles pharmacists should have in the healthcare system.

Clinical pharmacy services have been making inroads into practice in hospitals where pharmacists often provide cognitive services such as pharmaceutical care directly to patients. Hospital pharmacists still meet barriers limiting direct patient care in hospitals.

A pharmacist needs to have good rapport and connections with doctors (GPs and other medical specialists) and other healthcare personnel to provide pharmaceutical care.[24] Clinical pharmacy service often involves collaborative practice with doctors, especially for prescription drugs.[45] A model by McDonough and Doucette[46] describes the collaboration stages between doctors and pharmacists for successful collaboration, ranging from "Professional awareness" to "Commitment to the Collaborative Working Relationship." Although the collaborative practice has its advantages, pharmaceutical care could also be given by pharmacists without consulting other healthcare professionals as allowed by law in many countries. Changes in pharmacy education in some countries (e.g., the United States) are making pharmacists highly competent in the patient care process.[26,47] However, in some countries—such as the Nordic countries—the development has been much slower within academic institutions training pharmacists, and there exists a large gap between the visions for patient-centered care and a reality where pharmacy has a traditional focus on dispensing drug products.[48,49]

The American Pharmaceutical Association has set out prerequisites for introducing pharmaceutical care. They include an emphasis on establishing and maintaining a professional relationship; keeping patient-specific medical information; evaluating patient-specific medical information and developing a drug therapy plan with the patient; assuring that the patient has all information and knowledge necessary to carry out the drug therapy plan; and reviewing, monitoring, and modifying the therapeutic plan as necessary and appropriate, in concert with the patient and healthcare team.[50]

Reaching clinical consensus on pharmacotherapy and drug therapy changes is challenging. As per an article, the acceptance rate of pharmacists' interventions by other healthcare professionals ranges from 17% to 86% depending on the study design and care settings.[51] For pharmacists to increase the acceptance from doctors, they as professionals have to fulfill as many as possible of the following criteria. They need to

- have sufficient clinical experience,
- have evidence-based practice,
- have close and established collaboration with the doctor,
- have access to the patient's medical records,
- have access to quality drug information resources,
- be able to interview patients, and
- have patients referred for counseling with the pharmacist.

When doctor's involvement is needed, there should be a face-to-face communication between the doctor and the pharmacist about the drug therapy problems and possible solutions, and the pharmacists should be able to give clear suggestions for actions to be taken, and follow-up systems must be in place.

There are other societal pressures such as hospital accreditation, increased patient safety requirements, and demands of professional agencies and patient right movements for more rational use of drugs. These pressures may, in fact, support the establishment and growth of pharmaceutical care service. At the same time, these factors increase the demands on pharmacists to provide higher quality clinical services. As pharmacists become more accepted as an important profession in patient care, they are expected to meet quality standards and take responsibility for the care provided.

UNDERSTANDING THE ENVIRONMENT AND STRUCTURE OF HEALTHCARE

A pharmacist can either work independently with patients and other professionals or work in multiprofessional team caring for patients. It is paramount that the pharmacist understands how other professions define their role in patient care and where the roles with the pharmacist may overlap. Social pharmacy as an academic discipline can complement the clinical pharmacy discipline and vice versa.

One example: Clinical pharmacists have in some countries expanded their role as independent prescribers. But can pharmacists be trusted as independent prescribers? If the same professional diagnoses the disease and dispenses the drugs, chances are higher for financial bias. This fact triggered the split between the pharmacy profession and the medical profession in the middle ages. A review of the physicians' prescription by an independent pharmacist reassures the patient that the therapy is appropriate, and that the person prescribing the drug is not a financial beneficiary of the sale.

Another example: The pharmacist cannot assume that the other professions or patients know what pharmacist can add to the other profession's expertise. To succeed in interprofessional healthcare teams, pharmacists must be complementary, rather than replacing the needed services of other healthcare professions.[52]

Pharmacists need to better understand the system they work within and how to interact with the system—a field mostly researched in social pharmacy. Clinical pharmacists must work interprofessionally to improve rational drug use and patient safety. They must, however, strengthen their understanding of the environment and structure of healthcare to succeed in an extended role in patient care.

CONCLUSION

Pharmacy practice is a broader concept than clinical pharmacy in that it focuses both on the clinical work of pharmacists with patients and on the social environment within which pharmacists work. Globally, pharmacists increasingly provide more clinical patient-centered services, either in the community pharmacies or other healthcare institutions. Pharmacists are increasingly spending more time in direct patient care or for its preparation. It is important for pharmacists to have a common understanding of pharmaceutical care, and that they expand their role to include research on future clinical pharmacy services with high social and economic impacts. Recognition of pharmacists as a healthcare professional needs strategic approaches. Pharmacists need to engage with the health organizations, other healthcare professions and the patients with more clinical approaches. Social pharmacy provides this helicopter perspective on the healthcare sector and provides a deeper insight into how the actors in the sector think and act.